

Roll No.

TCE-301

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

MECHANICS OF FLUIDS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Derive the expression for surface tension of a liquid. (CO1)

OR

- (b) A flat plate 0.30 m^2 in area moves edgewise through oil between large fixed parallel plates 10 cm apart. If the velocity of the plate is 0.60 m/s and the oil has a kinetic viscosity of $0.45 \times 10^{-4} \text{ m}^2/\text{s}$ and specific gravity 0.8, calculate the drag force when the plate is equidistant from both plates. (CO1)
2. (a) Explain in detail, the various steps used in Rayleigh's method of dimensional analysis. (CO1)

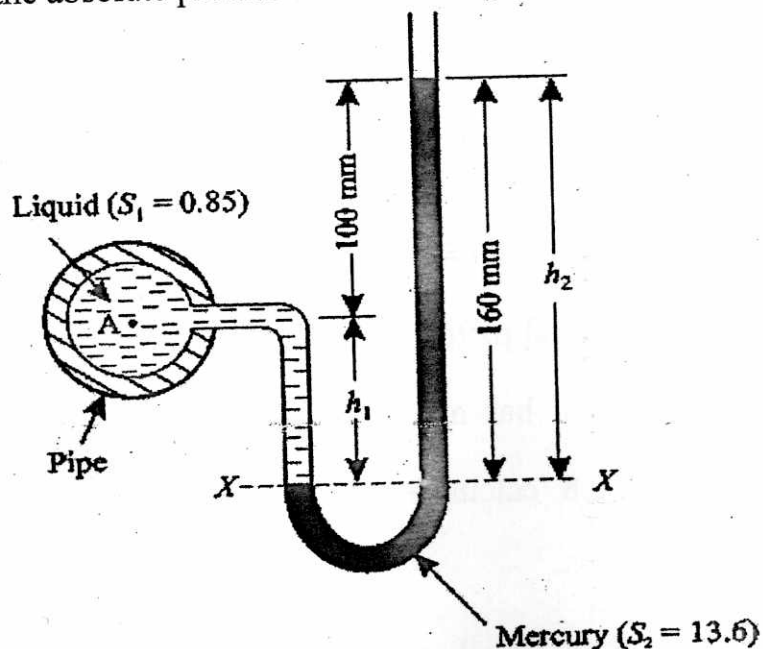
P. T. O.

OR

- (b) A pipe of diameter 1.50 m is required to transport an oil of specific gravity 0.90 and viscosity 3×10^{-2} poise at the rate of 3000 liter/sec. Tests were conducted on a 15 cm diameter pipe using water at 20°C. Find the velocity and rate of flow in the model. Take Viscosity of water at 20°C = 0.01 poise. (CO1)
3. (a) Derive the basic differential equation for the various of pressure in a fluid at rest. (CO2)

OR

- (b) A U-tube manometer is used to measure the pressure of oil of specific gravity 0.85 flowing in a pipe line. Its left end is connected to the pipe and the right-limb is open to the atmosphere. The centre of the pipe is 100 mm below the level of mercury (specific gravity = 13.6) in the right limb. If the difference of mercury level in the two limbs is 160 mm, determine the absolute pressure of oil in the pipe.

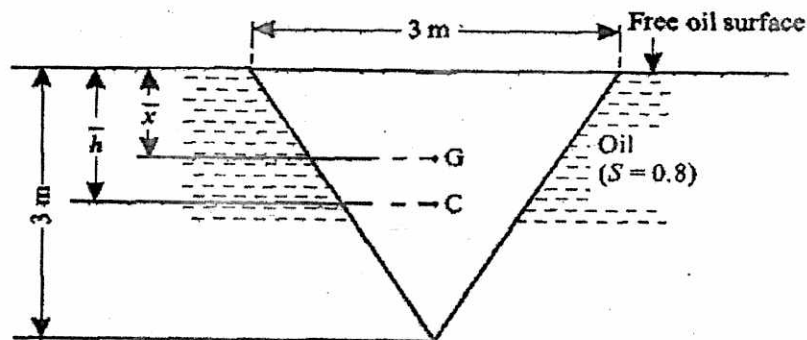


4. (a) Derive expressions for total pressure and centre of pressure for a vertically immersed surface. (CO2)

(3)

OR

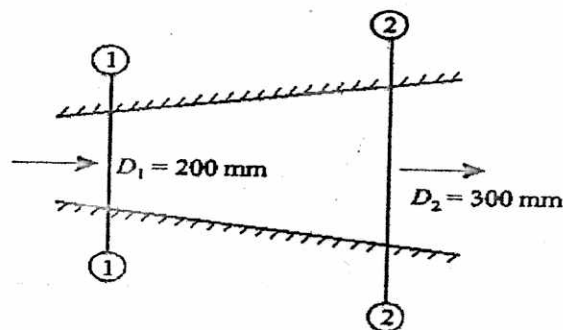
- (b) An isosceles triangular plate of base 3 and altitude 3 m is immersed vertically in an oil of specific gravity 0.8. The base of the plate coincides with the surface of oil. Determine : Total pressure on the plate and Centre of pressure. (CO2)



5. (a) What do you mean by Flow Net ? Describe the use and limitations of the flow net. (CO2)

OR

- (b) The diameter of a pipe at the sections 1-1 and 2-2 are 200 mm and 300 mm respectively. If the velocity of water flowing through the pipe at section 1-1 is 4 m/s, find : (CO2)
- (i) Discharge through the pipe, and
 - (ii) Velocity of water at section 2-2.



Roll No.

TCE-302

B. TECH. (CE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

BASIC SURVEYING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define Surveying and explain the basic principles of surveying with the help of neat diagram. (CO1)

OR

- (b) Difference between Geodetic survey and Plane surveying. Classify different types of survey on the basis of instruments used. (CO1)

2. (a) Explain any *two* of the following terms related to chain surveying with the help of neat diagram : (CO1)

Line ranger, Cross staff, Optical Square, Clinometer.

OR

- (b) A base line is measured with a steel tape, it is approximately 1000 m long. Calculate the correct length of the baseline at MSL when the pull at the standardization equal 25 kg. The pull applied is 32 kg. Cross

P. T. O.

sectional area of the tape is 0.0645 cm^2 , $E = 2.11 \times 10^6 \text{ kg/cm}^2$. Temperature at the time of measurement was 35°C and standardized at 20°C respectively. The difference of level between the two ends of the base line is 2.0 m , Radius of earth $R = 6370 \text{ km}$. Elevation of base line above MSL is 1000 m , $\alpha = 12 \times 10^{-6}/^\circ \text{C}$. Determine the correct length of line after applying the necessary corrections as per the data given in the numerical. (CO1)

3. (a) Explain the following terms related to surveying with the help of neat diagram : (CO1)
- (i) Control points
 - (ii) Agonic lines
 - (iii) Traversing
 - (iv) Ranging
 - (v) Base line

OR

- (b) The area of the plan of an old survey plotted to a scale of $10 \text{ metres to } 1 \text{ cm}$ measures now as 100.2 sq. cm as found by a planimeter. The plan is found to have shrunk so that a line originally 10 cm long now measures 9.7 cm only. (CO1)

Find :

- (i) Shrunk scale
 - (ii) True area of the survey.
4. (a) Explain difference between Prismatic compass and Surveyor's compass with a sketch. (CO2)

(3)

OR

- (b) The following bearings were observed in a compass traverse over an area where local attraction was suspected. Find the amount of local attraction at different stations, the correct bearing of the line and the included angles. Also draw the sketch on a scale AB=100 m, BC= 100 m and CD= 50 m : (CO2)

Line	F. B.	B. B.
AB	$68^{\circ}15'$	$248^{\circ}15'$
BC	$148^{\circ}45'$	$326^{\circ}15'$
CD	$224^{\circ}30'$	$46^{\circ}00'$
DE	$217^{\circ}15'$	$38^{\circ}15'$
EA	$237^{\circ}45'$	$147^{\circ}45'$

5. (a) Explain step by step procedure of measuring the angles by repetition method. Enlist the errors that are removed by this method. (CO2)

OR

- (b) The fore bearings of lines AB, BC, CD and DE, are $45^{\circ} 30'$, $120^{\circ} 15'$, $200^{\circ} 30'$ and $280^{\circ} 45'$. Find angle B, C and D. (CO2)

Roll No.

TCE-303

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

BUILDING MATERIALS AND CONSTRUCTION TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What are the constituents of good brick earth ? (CO1)

OR

(b) Classify the types of bricks based on size and shape. Elaborate the function of frog in red clay bricks. (CO1)

2. (a) What is seasoning of timber ? State various methods of seasoning and explain any *one* of them. (CO1)

OR

(b) Define timber and explain the cross-section of trunk of timber with the help of a neat diagram. (CO1)

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(2)

3. (a) Discuss the operation of preparation of clay for the manufacture of bricks. (CO1)

OR

- (b) Define mortar and elaborate the various types of mortar used. (CO1)

4. (a) What are admixtures ? Briefly explain its uses. (CO2)

OR

- (b) Enlist the various laboratory tests of cement. Explain any *two*. (CO2)

5. (a) Describe the field tests for the cement. (CO2)

OR

- (b) Define cement and explain the composition of ordinary cement. (CO2)

Roll No.

TCE-304

**B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

STRENGTH OF MATERIALS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the following terms :

(i) Young's Modulus of Elasticity

(ii) Poisson's Ratio

(iii) Hooke's Law

(CO1)

OR

(b) A concrete column of cross-sectional area of 350×350 mm is reinforced by four longitudinal 30 mm diameter round steel bars placed at each corner. If the column carries comprehensive load of 400 kN, determine comprehensive stresses produced in the concrete and steel bars. Assume that Young's modulus of elasticity of steel is 15 times that of concrete.

(CO1)

P. T. O.

(2)

2. (a) Compare the moment of resistance of circular and square sections of same materials and that of same cross section area. (CO1)

OR

- (b) Deduce an expression for Pure Bending of beam equations. (CO1)
3. (a) A cantilever beam of span 6 m is carries a concentrated load of 20 kN at the center of the span. Calculate the reaction and moments at fixed end and draw the SF and BM diagrams. (CO2)

OR

- (b) A rectangular beam 60 mm wide and 150 mm deep is simply supported over a span of 4 meters. If the beam is subjected to a uniformly distributed load of 4.5 kN/m, find the maximum bending stress induced in the beam. (CO2)
4. (a) Distinguish between Direct Stress and Bending Stress. (CO2)

OR

- (b) Derive the expression for shear stress distribution over I-section. (CO2)
5. (a) Deduce on expression for equation of pure bending with it's all assumptions. (CO2)

OR

- (b) A simply supported beam of 4 m span carries a uniformly distributed load of 100 kN per m over the whole span. The cross section of the beam is a T section with a flange width of 100 mm web and flange thickness of 20 mm and overall depth of 140 mm. Determine the maximum shear stress in the beam and draw the stress distribution for the section. (CO2)

Roll No.

TCE-305

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

ENGINEERING MECHANICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

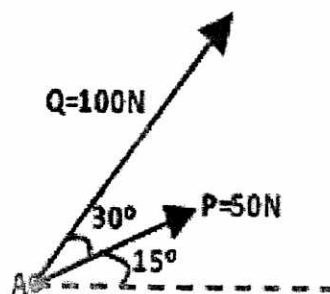
(ii) Each sub-question carries 10 marks.

1. (a) State and explain the Parallelogram Law of Forces. (CO1)

OR

- (b) What is free body diagram ? Explain with an example. Also, list out the assumption made while drawing a free body diagram. (CO1)

2. (a) Two forces are acting at a point as shown in the figure below. Determine the magnitude and direction of the resultant. (CO1)



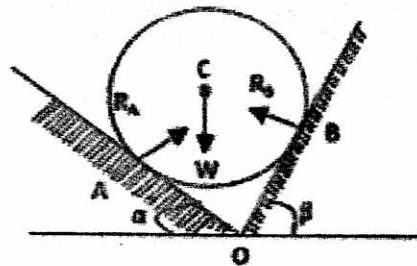
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TCE-305

OR

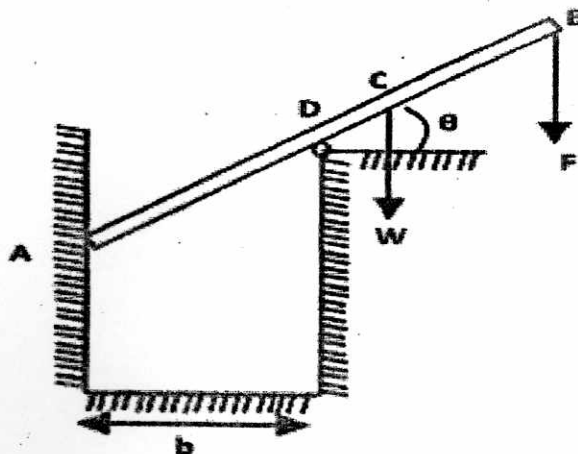
- (b) A smooth circular cylinder of weight W and radius r rests in a V-shaped groove whose sides are inclined at angle α and β to the horizontal as shown in fig. Find the reaction R_A and R_B at the point of contact. Given $\alpha = 30^\circ$, $\beta = 60^\circ$ and $W = 1000 \text{ N}$. (CO1)



3. (a) A body of weight W is resting on a smooth inclined plane. Draw the FBD of the body. Also write the equation of equilibrium. (CO2)

OR

- (b) A prismatic bar AB of weight W is resting against a smooth wall at A and is supported on a small roller at point D. If a vertical force F is applied at the end B, find the position of equilibrium as defined by the angle θ . Assume $AB = l$. (CO2)



(3)

4. (a) Define the following :

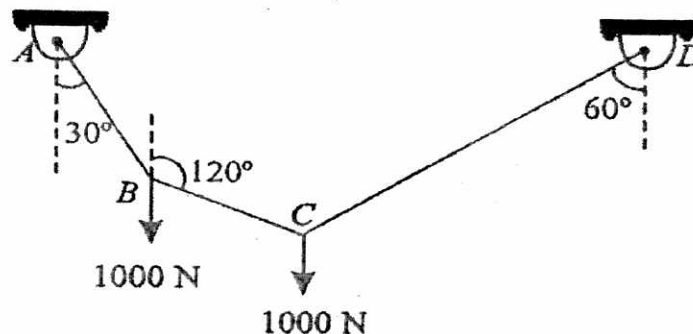
(i) Varignon's principle of moments

(ii) Lami's Theorem (With proof)

(CO2)

OR

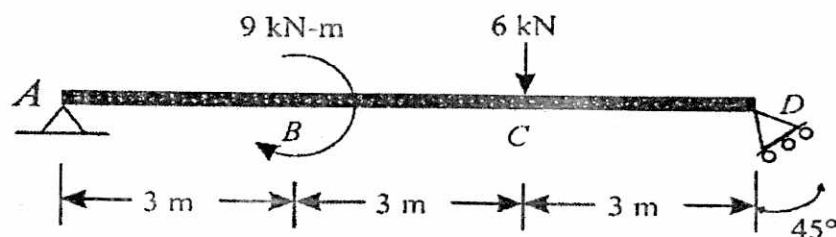
(b) A string ABCD, attached to fixed points A and D has two equal weights of 1000 N attached to it at B and C. The weights rest with the portions AB and CD inclined at angles as shown in Figure below. Find the tensions in the portions AB, BC and CD of the string, if the inclination of the portion BC with the vertical is 120° . (CO2)



5. (a) Describe the various types of beams with the help of diagrams. (CO3)

OR

(b) Find the reactions at supports on a beam carrying various loads as shown in the figure below : (CO3)



Roll No.

TMA-302

B. TECH. (CIVIL ENGINEERING) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

ENGINEERING MATHEMATICS-III

Time : 1:30 Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Prove that : (CO1)

$$f(z) = 2x - x^3 + 3xy^2$$

is harmonic function and find its harmonic conjugate. Also find the corresponding analytic function. (CO1)

OR

- (b) Find a real root of the equation $xe^x = 1$ by using Newton-Raphson method correct upto 3 decimal places. (CO2)

2. (a) Evaluate the integral $\int_{(1, -1)}^{(2, 1)} (2x + iy + 1) dz$ along the straight line from $z = 1 - i$ to $z = 2 + i$. (CO1)

P. T. O.

OR

- (b) Using Bisection method, to find a real of an equation $x \log_{10} x = 1.2$ correct to three decimal places. (CO2)

3. (a) Using Trapezoidal rule, find the value of the integral $\int_4^{5.2} \log x dx$ (Accuracy correct to three decimal places). (CO3)

OR

- (b) Check the analyticity of the following functions : (CO1)

(i) $f(z) = |z|^{\frac{1}{2}}$ for all z .

(ii) $f(z) = z^2$ for all z .

4. (a) Show that the function : (CO1)

$$f(z) = u + iv$$

where

$$f(z) = \begin{cases} \frac{x^2 y^5 (x + iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

is not analytic at $z = 0$ although Cauchy-Riemann equations are satisfied at $z = 0$.

OR

- (b) Define analytic function. Show that the function :

$$f(z) = z|z|$$

is not analytic anywhere. (CO1)

(3)

5. (a) Using Cauchy's integral formula compute the value of the integral :

(CO1)

$$\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$$

where

$$C : |z| = 3.$$

OR

- (b) Show that :

$$u(x, y) = \sin x \cosh y + 2 \cos x \sinh y + x^2 - y^2 + 4xy$$

is harmonic. Find the analytic function $f(z)$ with u as its real part by using Milne-Thomson's method.

(CO1)